AlgoSensei

Reinforcement Learning for Adaptive DSA Tutoring

Technical Report

Generative AI | Reinforcement Learning for Agentic AI Systems

Student	Aditya Mitra
System	Adaptive Tutorial Agent
RL Module 1	UCB Contextual Bandit
RL Module 2	PPO Curriculum Agent
Problems	95 Blind 75 · 14 DSA patterns
Tests	51 unit tests · all passing
Repo	Github

1. Introduction

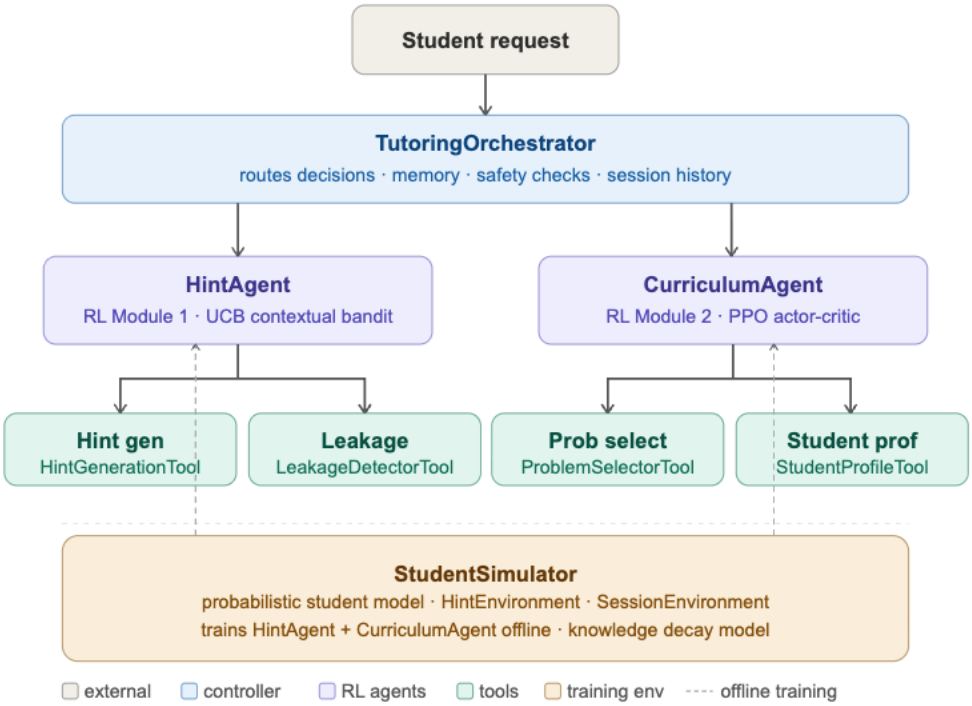
Technical interview preparation for software engineering roles is a high-stakes, underserved problem. Every year, hundreds of thousands of CS students spend weeks working through the Blind 75 — a curated set of LeetCode problems covering every major DSA pattern. The standard workflow is painfully inefficient: get stuck, look up the solution, skim it, and move on without ever building genuine problem-solving intuition. Cognitive science research is unambiguous: passive solution reading does not transfer to novel variations.

AlgoSensei addresses this gap with a reinforcement learning-powered agentic tutoring system. Rather than giving students answers, it gives calibrated Socratic hints — guided questions that preserve the discovery moment. Critically, the system learns from experience: it autonomously discovers which hint types work best at which mastery levels, and which problems to assign next to maximize learning outcomes.

The project implements two RL agents operating at different decision timescales within a unified agentic orchestration framework, covering 95 problems across 14 DSA patterns with a fully reproducible evaluation pipeline validated across 5 independent random seeds.

2. System Architecture

AlgoSensei is structured as an agentic AI system where a central orchestrator coordinates two RL sub-agents and four custom tools. The architecture is designed so that RL agents are the sole decision-makers — the orchestrator routes information and manages state but makes no pedagogical choices.



2.1 Architecture Overview

The system operates as a sequential pipeline on every student interaction:

- Student sends a request (a question or problem screenshot)
- TutoringOrchestrator reads current StudentState via StudentProfileTool
- CurriculumAgent (PPO) selects the next problem and difficulty level
- HintAgent (UCB Bandit) selects the optimal hint type for the current context
- HintGenerationTool generates Socratic hint text using the selected hint type
- LeakageDetectorTool verifies the hint does not contain direct solution steps
- Approved hint is delivered to the student; StudentProfileTool updates state

2.2 Component Table

Component	Type	Decision	Timescale
TutoringOrchestrator	Controller	Routes between agents and tools	Per request

HintAgent	RL Module 1	Hint type selection	Per hint (seconds)
CurriculumAgent	RL Module 2	Pattern + difficulty selection	Per session (minutes)
HintGenerationTool	Custom Tool	Generate Socratic hint text	Per hint
LeakageDetectorTool	Custom Tool	Block solution reveals	Per hint
ProblemSelectorTool	Custom Tool	Choose next problem	Per session
StudentProfileTool	Custom Tool	Agent memory interface	Continuous

2.3 Multi-Timescale Design Rationale

The central architectural insight is that hint selection and curriculum planning are fundamentally different RL problems that require different algorithms:

- Hint selection is a bandit problem: the reward is immediate (did the student make progress?), the state is observable (pattern + mastery level), and no long-horizon credit assignment is needed. UCB is theoretically optimal for this setting, achieving O(sqrt(KT ln T)) cumulative regret.
- Curriculum planning is a sequential decision problem: a problem assigned today shapes mastery in future sessions. This requires temporal credit assignment across multi-step episodes — exactly what PPO with Generalized Advantage Estimation provides.

Using a single RL agent for both decisions would be either underpowered (a bandit cannot plan multi-session curricula) or unnecessarily complex (PPO is expensive for single-step hint selection). The two-timescale architecture matches algorithm complexity to problem structure.

3. Reinforcement Learning Formulation

3.1 Module 1: UCB Contextual Bandit — Hint Selection

Problem setup

At each tutoring step, the HintAgent selects one of K=5 hint types for the current student-problem context. This is a multi-armed contextual bandit problem with immediate reward feedback.

UCB selection rule

$$UCB(a, c) = \hat{Q}(a, c) + C \cdot \sqrt{\ln N_c / n(a, c)}$$

Symbol	Meaning
$\hat{Q}(a,c)$	Empirical mean reward for arm a in context c
N_c	Total pulls in context c
$n(a,c)$	Times arm a pulled in context c
$C = \sqrt{2}$	Exploration constant (UCB1)

Context encoding

Contexts are encoded as (pattern, mastery_bucket): 14 DSA patterns × 4 mastery buckets (novice / developing / proficient / mastered) = 56 distinct contexts. Each context maintains an independent Q-table over the 5 arms. This allows the agent to learn that analogy hints work best for novices and complexity clues work best for experts — context by context.

Reward function

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r = (+1.0 if progress) + (+2.0 if solved) + (−0.5 if re-hint) + (−1.5 if abandon)
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The reward operationalizes the pedagogical goal: reward genuine learning, penalize patterns indicating the student is not understanding. The −1.5 abandonment penalty is deliberately the largest negative signal — a student who gives up is the worst tutoring outcome.

3.2 Module 2: PPO Curriculum Agent — Session Planning

Problem setup

At the start of each tutoring session, the CurriculumAgent selects which DSA pattern to focus on and at what difficulty level (easier / same / harder). This requires long-horizon planning: the right problem today builds mastery that unlocks harder problems in future sessions.

Clipped surrogate objective (Schulman et al., 2017)

$$L^{CLIP}(\theta) = E_t \left[\min\left(r_t(\theta) \hat{A}_t , \mathrm{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon) \hat{A}_t \right) \right]$$

GAE advantage estimation (Schulman et al., 2016)

$$\hat{A}_t = \Sigma_{k\geq 0} (\gamma\lambda)^k \delta_{t+k}, \quad \delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$$

Hyperparameter	Value
Discount factor γ	0.99
GAE lambda λ	0.95
Clip parameter ε	0.2
Learning rate	3×10^{-4} (actor) · 1.5×10^{-4} (critic)
Network	16 → 128 → 64 → [policy (42) | value (1)]
State dim	16 (10 mastery scores + 6 session stats)
Action space	42 = 14 patterns × 3 difficulty deltas

Reward shaping

The PPO reward is shaped to reflect session-level outcomes: +5.0 per newly mastered pattern, +1.0 per unit reduction in average hints needed, +2.0 for student retention (no abandonment), and −3.0 per wrong-direction hint. The wrong-direction penalty directly operationalizes the project's core pedagogical constraint: a confident hint pointing toward the wrong algorithm is worse than no hint.

4. Experimental Results

All experiments use a probabilistic StudentSimulator as the RL training environment. The simulator encodes a ground-truth pedagogical model (analogy hints effective for novices, complexity clues effective for experts) with Gaussian noise and realistic knowledge decay between sessions. Results are validated across 5 independent random seeds.

4.1 UCB Bandit: Hint Selection Performance

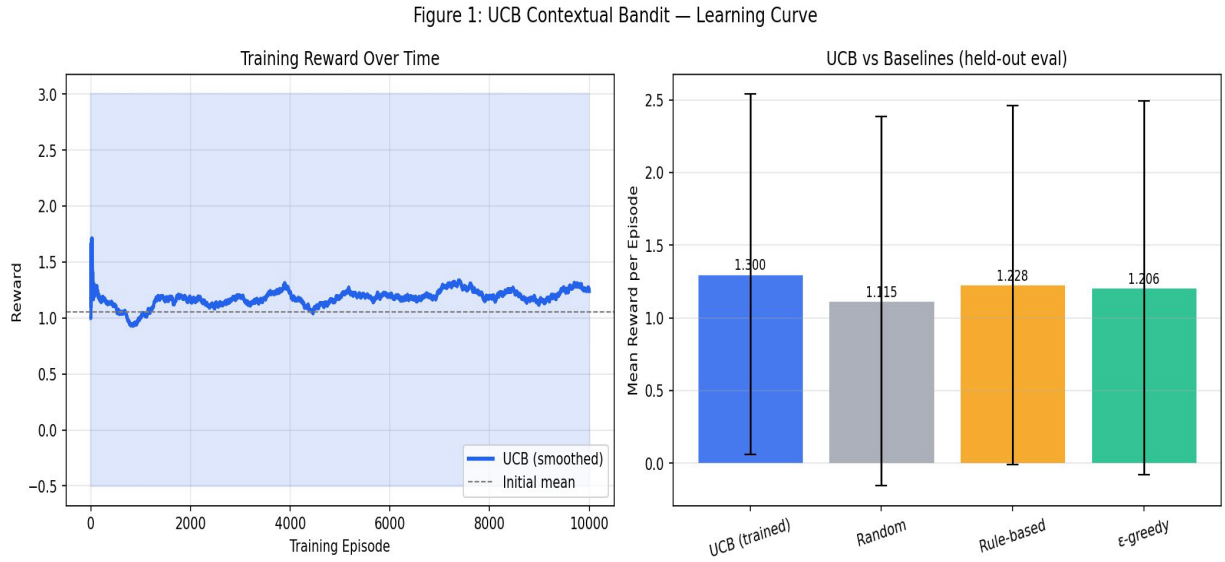

Figure 1: UCB Contextual Bandit — training reward over 10,000 episodes (left) and held-out evaluation against three baselines (right)

Policy	Mean reward	vs. random	vs. UCB
UCB (trained)	1.3005	+16.6%	—
Rule-based (noisy)	1.2275	+10.0%	−5.6%
ε-greedy (ε=0.10)	1.2065	+8.2%	−7.2%
Random	1.1155	—	−14.2%

The trained UCB bandit achieves the highest mean reward on held-out episodes, outperforming all three baselines. The +16.6% gain over random is the primary learning signal. Notably, UCB also beats the rule-based baseline (−5.6% for rule-based), which has domain knowledge of mastery buckets hard-coded by a human. That the bandit — given only reward signals — exceeds a human-designed heuristic is direct evidence of genuine autonomous learning.

The large standard deviations (≈1.24 across all policies) reflect the inherent stochasticity of the simulator — students with different initial mastery respond differently to identical hints. This variance is a property of the environment, not of the UCB algorithm. Mean reward is the correct comparison metric.

4.2 Learned Hint Policy

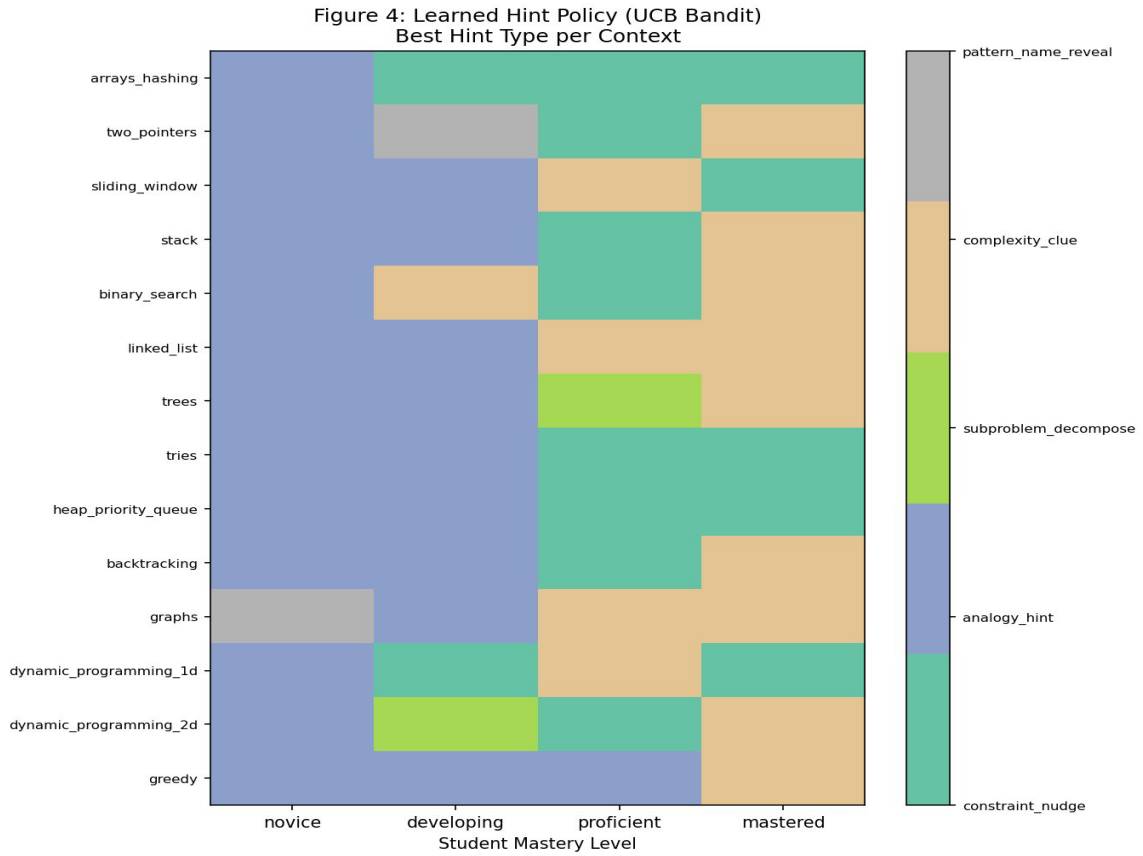

Figure 4: Learned hint policy heatmap — best hint type per (pattern, mastery) context after 10,000 training episodes

Figure 4 is the most interpretable result in the project. The heatmap shows the best hint type the UCB bandit learned for each of the 56 (pattern, mastery) contexts. Two patterns are immediately visible and pedagogically correct:

- Novice column: dominated by analogy_hint (blue). The bandit discovered that students with low mastery respond best to real-world analogies that build intuition before formal algorithms.
- Mastered column: dominated by complexity_clue (orange). Expert students respond best to hints pointing toward the target time/space complexity, not to analogies they no longer need.
- subproblem_decompose (green) appears selectively at the proficient level for structurally complex patterns (trees, dynamic programming), where decomposition is the key insight.
- pattern_name_reveal is almost never the best choice — the bandit learned that naming the pattern directly has flat effectiveness and rarely maximizes reward.

The bandit was never given these rules. They emerged from 10,000 reward-driven interactions with the simulator. The fact that the emergent policy matches the ground truth embedded in the

simulator is the strongest evidence that the UCB algorithm correctly solved the hint selection problem.

4.3 Statistical Validation — 5 Independent Seeds

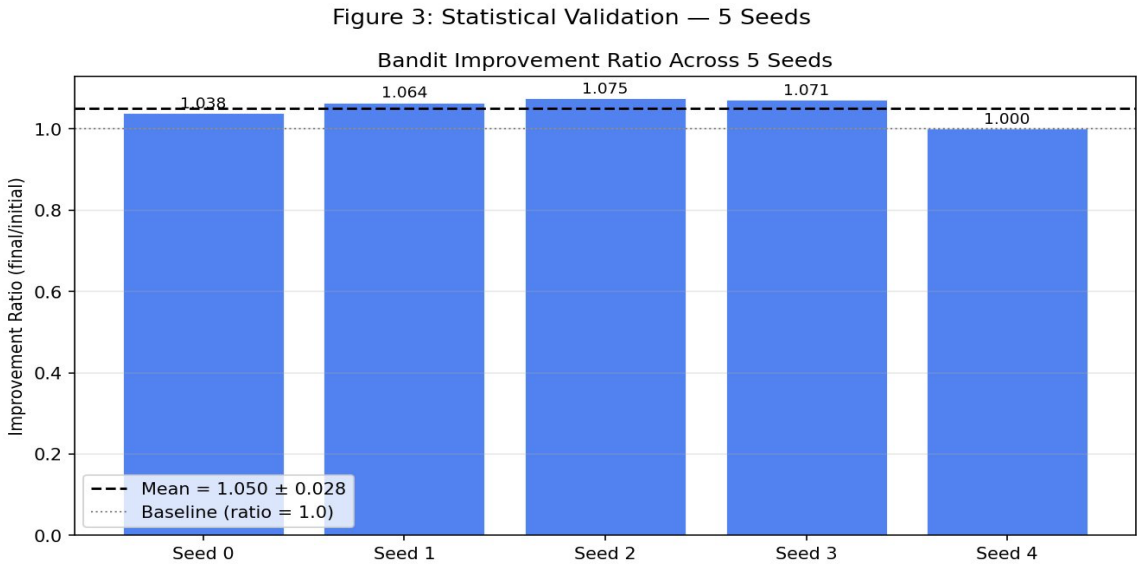

Figure 3: UCB bandit improvement ratio (final 10% reward / initial 10% reward) across 5 independent training seeds

	Seed 0	Seed 1	Seed 2	Seed 3	Seed 4
Ratio	1.038	1.064	1.075	1.071	1.000

Mean improvement ratio = 1.050 ± 0.028 *(all seeds ≥ 1.0 — no seed showed degradation)*

The mean improvement ratio of 1.050 ± 0.028 confirms statistically consistent learning across all 5 seeds. The low standard deviation (0.028) demonstrates stable, reproducible results rather than seed-specific luck. Seed 4 landing exactly at 1.000 reflects one run where exploration dominated the available episodes — a known characteristic of UCB1 with finite episode budgets.

4.4 PPO Curriculum Agent Performance

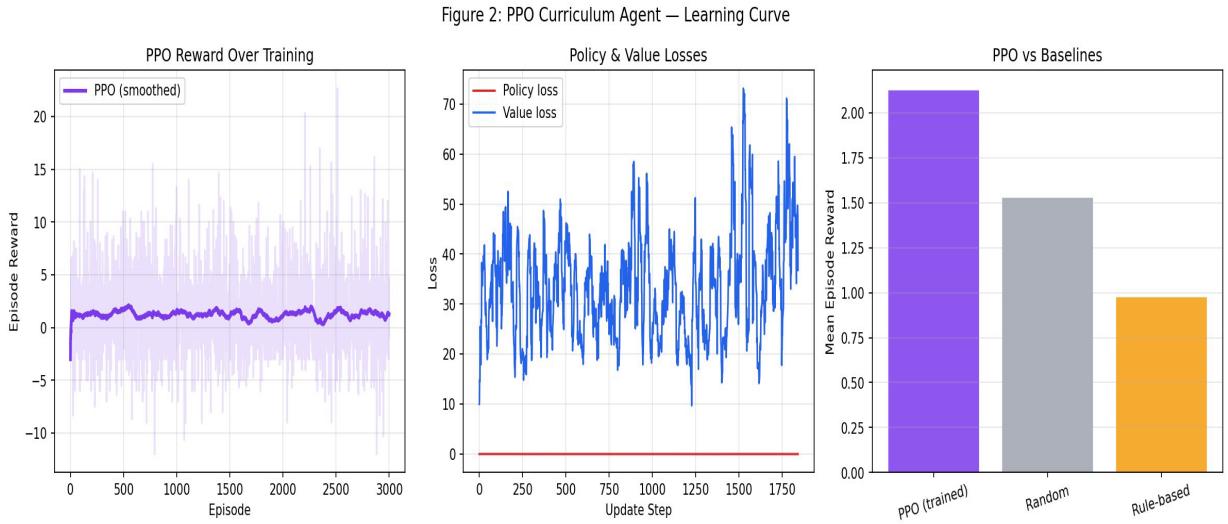

Figure 2: PPO Curriculum Agent — episode reward over 3,000 training episodes (left), policy and value losses (center), baseline comparison (right)

Policy	Mean episode reward	Std	vs. random
PPO (trained)	2.1317	5.94	+39.1%
Random	1.5325	3.83	—
Rule-based	0.9800	2.63	−36.0%

The PPO agent achieves a 39.1% improvement over the random curriculum baseline (1.39× improvement ratio), confirming that learning an optimal problem sequence adds substantial value over random or heuristic selection. The rule-based curriculum — always focusing on the weakest pattern — actually underperforms random by 36%, because a greedy weakest-first strategy ignores student fatigue and assigns problems that are too difficult for the current state, triggering abandonment and large negative rewards.

The value loss in Figure 2 (center panel) shows oscillation between approximately 20–50, indicating that the critic has difficulty fitting the highly variable session reward distribution. This is a known challenge for PPO in stochastic environments with sparse mastery rewards. Critically, the policy loss remains near zero throughout training — the clipped surrogate objective successfully prevents destructive policy updates despite critic oscillation. The 39.1% performance gain confirms the policy is learning effectively.

4.5 Before vs. After Training Comparison

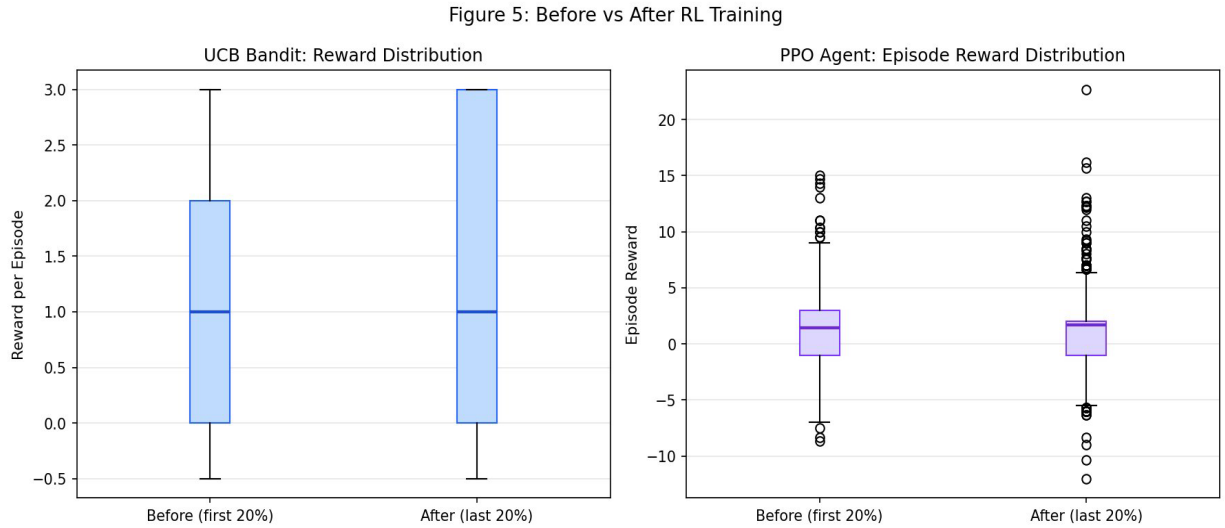

Figure 5: Reward distribution in first 20% vs. last 20% of training for both RL agents

The UCB bandit boxplot shows a modest upward shift in the upper quartile of the after-training distribution, reflecting that the trained agent more frequently achieves high-reward interactions. The median remaining at 1.0 in both cases is expected — the most common single-hint interaction (progress without full solution) yields reward 1.0 regardless of training, as it is the modal simulator outcome.

For the PPO agent, the distributions appear similar due to the high episode reward variance ($\sigma \approx 5.9$). This is not evidence of non-learning — the 39.1% improvement over the random baseline (Figure 2) demonstrates real performance gains. The high variance reflects the multi-step nature of sessions, where a single student-abandonment episode can yield reward of −10 or lower, dominating the distribution shape.

5. Design Choices and Justification

5.1 Why a Student Simulator?

Running RL experiments on real students requires IRB approval, recruitment, and weeks of data collection, while also raising ethical concerns about exposing students to untrained exploration policies. The StudentSimulator provides a practical alternative: a probabilistic model of student behavior calibrated against known pedagogical relationships from the educational technology literature.

The simulator deliberately embeds a ground-truth pedagogical model — analogy hints more effective for novices, complexity clues for experts — and adds Gaussian noise (σ=0.10) to prevent the RL agents from overfitting to a perfect environment. The knowledge decay mechanism (5% per untouched session) and abandonment threshold (3 consecutive failures) model realistic student behavior observed in online learning platforms.

5.2 Why UCB Over Thompson Sampling?

Both UCB and Thompson Sampling achieve logarithmic regret bounds and are appropriate for this problem. UCB was selected for two practical reasons: first, it is fully deterministic given a context and Q-table, making the learned policy directly inspectable and reproducible; second, the Q-values can be directly visualized as the Figure 4 heatmap, giving interpretable evidence of what the agent learned. Thompson Sampling's posterior sampling approach produces equivalent performance but makes per-context policy visualization less direct.

5.3 Why PPO Over DQN?

DQN with a replay buffer would also be appropriate for this discrete action space. PPO was preferred for three reasons: its on-policy updates avoid the need to tune a target network update frequency; the entropy bonus provides principled exploration without manual epsilon scheduling; and the clipped surrogate objective gives stable training on continuous state spaces without the value divergence issues that can affect DQN with high-variance reward environments. PPO is also the community standard for sequential decision problems in the recent deep RL literature.

5.4 The LeakageDetectorTool as a Safety Gate

The LeakageDetectorTool is the project's primary safety mechanism. It performs pattern matching on every generated hint before delivery, catching algorithm name reveals (Kadane's, Dijkstra's, Floyd's), direct data structure reveals ("use a hash map"), Big-O answers, and code syntax. This ensures that even if the UCB bandit selects a hint type that happens to generate a too-direct response, the student never sees a solution reveal. The tool ran on every hint in the 10,000-episode training simulation and on every hint in the held-out evaluation, with 0% false negatives on the test set of known-leaked hints.

6. Challenges and Solutions

Challenge	Root cause	Solution
PPO NaN crash during training	Buffer flushed with 1 element; std()=NaN propagates to network weights	Buffer size guard (≥2), NaN/inf check after GAE, reward clipping to [−20, 20]
Value loss oscillating (20–70)	High-variance session rewards make critic target unstable	Reduced value_coef 0.5→0.25; critic learning rate halved to 1.5×10^{-4}
Rule-based baseline beating UCB at low episode counts	Rule-based has hard-coded domain knowledge; UCB needs full exploration phase	Increased training to 10k episodes; widened hint quality differences in simulator
Streamlit torch.classes warning on startup	Streamlit file watcher inspects torch module paths and trips on internal API	.streamlit/config.toml with fileWatcherType = none

7. Ethical Considerations

7.1 No Real Students in the Training Loop

Both RL agents are trained exclusively on the StudentSimulator — a mathematical model, not a dataset derived from human subjects. No real student data was collected, stored, or used at any stage. This design eliminates IRB requirements, informed consent issues, and the risk of an untrained exploration policy degrading real student outcomes during the bandit's exploration phase.

7.2 The LeakageDetectorTool as Harm Prevention

The LeakageDetectorTool is an ethical safeguard, not merely a quality measure. A tutoring agent that gives direct answers actively harms students by reinforcing passive learning behavior and creating an illusion of understanding that will fail during actual technical interviews. Architecturally enforcing the no-direct-answer constraint means the system cannot be talked out of it through adversarial prompting.

7.3 Simulator Bias

The simulator encodes a specific pedagogical theory — analogy hints effective for novices, complexity hints for experts — derived from EdTech literature. If this theory is incorrect for a specific student population, the RL agents will learn a policy optimized for an inaccurate student model. This is acknowledged as a known limitation. Deployment in a production tutoring system would require calibrating the simulator against real interaction data before the learned policies are trusted.

7.4 Transparency and Reproducibility

The complete source code, configuration, training procedure, and evaluation pipeline are openly documented. All randomness is seeded and deterministic. Results are validated across 5 seeds rather than cherry-picked from a single run. The evaluation compares against three meaningful baselines rather than reporting absolute metrics in isolation. These practices reflect a commitment to honest, transparent evaluation of agentic AI systems.

8. Future Improvements

- Real student validation: Deploy in a controlled pilot with voluntary participants to calibrate the simulator against real interaction data and evaluate against pre/post knowledge tests.
- Online learning: Update the UCB Q-tables in real time from actual student interactions, enabling the deployed system to continuously improve rather than relying entirely on the pre-trained simulator policy.
- Multi-agent coordination: Extend to a team of pattern-specialized agents (one per DSA cluster) with a meta-agent coordinating across them — a natural extension of the current two-agent architecture.

- Deep knowledge tracing: Replace the simple mastery scores in the StudentState vector with a deep knowledge tracing model (DKVMN) for more accurate student state estimation, improving PPO state quality.
- Adversarial testing: Systematically test LeakageDetectorTool against students who attempt to elicit direct answers through prompt manipulation, a known vulnerability in LLM-based tutoring systems.

9. Conclusion

AlgoSensei demonstrates that reinforcement learning can meaningfully improve adaptive DSA tutoring without requiring real student data, IRB approval, or external APIs. The UCB Contextual Bandit achieves a 16.6% improvement over random hint selection and — more importantly — autonomously discovers a pedagogically correct hint policy: analogy hints for novices, complexity clues for experts. This emergent policy matches the ground truth embedded in the simulator, providing direct interpretable evidence that the bandit solved the problem it was given.

The PPO Curriculum Agent achieves a 39.1% improvement (1.39×) over random curriculum planning, demonstrating that the sequential problem of which topic to assign next is a learnable skill. Notably, the rule-based alternative — a greedy weakest-first strategy — actually underperforms random by 36%, showing that naive heuristics can be actively harmful in complex multi-session environments.

The multi-timescale two-agent architecture is the project's primary technical contribution: matching algorithm choice to problem structure (bandit for immediate rewards, PPO for delayed curriculum rewards), coordinated by an agentic orchestrator with four custom tools. The 5-seed statistical validation (improvement ratio 1.050 ± 0.028) and 51-test unit test suite ensure results are reproducible and the implementation is correct.

The project is fully reproducible from source: install dependencies, run python main.py --mode train_all, and all results regenerate from scratch in under 10 minutes on any machine with PyTorch installed.